

Biraj Acherjee

Software Developer | Full Stack Developer | AI Enthusiast

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PROFESSIONAL SUMMARY

Software Engineer specializing in full-stack web development and AI-driven applications. Experienced in architecting end-to-end solutions using Python, React, and PostgreSQL, focusing on scalable backend systems and responsive user interfaces. Proven track record of developing complex platforms, including forensic intelligence systems and healthcare routing applications, leveraging NLP, knowledge graphs, and real-time mapping integrations.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL

Frontend: React, TypeScript, HTML5, CSS3, Tailwind CSS

Backend: Flask, FastAPI

Databases: PostgreSQL, Neo4j, SQLite, Redis

Tools: Git, GitHub, Postman, Linux, VS Code

Concepts: Data Structures & Algorithms, Object-Oriented Programming, REST APIs, Knowledge Graphs

PROJECTS

Tatva — AI Forensic Intelligence

- Architected an AI-powered forensic investigation platform with a FastAPI backend to ingest, process, and analyze heterogeneous digital evidence.
- Designed and implemented a knowledge graph ecosystem using Neo4j to map complex entity relationships and uncover hidden investigative patterns.
- Integrated Google Gemini API and SpaCy NLP pipelines for automated text extraction, named entity recognition, and forensic dossier generation.
- Built a responsive React and TypeScript frontend featuring interactive graph visualizations, backed by PostgreSQL for structured data persistence.

ClearRoute — Emergency Healthcare Navigation Platform

- Developed a real-time emergency healthcare discovery and navigation platform designed to optimize routing to the nearest available medical facilities.
- Engineered route optimization and ETA estimation algorithms leveraging the OpenStreetMap ecosystem, Overpass API, and OSRM.
- Created an interactive map interface using Leaflet.js, incorporating a voice search feature implemented via the Web Speech API for accessibility.
- Designed a scalable REST API architecture using a Flask backend to coordinate asynchronous geospatial data requests and client-side rendering.

Smart Study Planner

- Built a productivity dashboard and intelligent study planning application using Python and Streamlit to manage academic workflows and deadlines.
- Implemented robust task management and session state handling to track user progress and generate personalized schedules based on input criteria.
- Engineered a lightweight SQLite database layer to persist user configurations, study patterns, and dynamically updated syllabus components.